

Study and Implementation of Constant Current-Constant Voltage(CC-CV) Charger for Lithium-ion(Li-ion) Battery using Machine Learning(ML)

V Devi Maheswaran

Rajalakshmi Engineering College
Electrical and Electronics Engineering
Chennai,India

devimaheswaran.v@rajalakshmi.edu.in

S B Ron Carter

Rajalakshmi Engineering College
Electrical and Electronics Engineering
Chennai,India

V. Vijay

Rajalakshmi Engineering College
Electrical and Electronics Engineering
Chennai,India

S Vasanth Raj

Rajalakshmi Engineering College
Electrical and Electronics Engineering
Chennai,India

Abstract— Battery charging techniques plays a vital role in electric mobility applications as an energy storage system. Lithium-ion batteries have become indispensable in portable devices, electric vehicles and solar powered devices. In order to maximize the potential of these rechargeable batteries, this paper focuses on the basics to advanced study of Lithium-ion battery charging techniques. The necessary safety aspects and implementation of the charging methodologies are also discussed. The Constant Current (CC) scheme charges with a low, constant current to obtain full charge only at the end. Constant Voltage (CV) scheme has to maintain a constant voltage in order to charge the batteries and prolong its life. Hence the objective of this work is to integrate both CC and CV charging circuit for a lithium-ion battery. To prolong battery lifespan and improve the safety aspects, step by step study of combined CC-CV charging circuit is presented. Initially, the CC method is utilized to charge the battery until it reaches the specified State of Charge (SoC). This effectively eliminates overcurrent situation. Next, the CV method is involved to charge the remaining battery capacity, guaranteeing an efficiency of nearly 90%. Further, the focus on implementing machine learning algorithms to charging circuits are also discussed.

Keywords—Battery, Li-ion, Constant Current, Constant Voltage, Charging, Discharging

I. INTRODUCTION

Lithium-ion (Li-ion) batteries are essential in modern electronics and electric vehicles due to their high energy density and lightweight properties. Maximizing their performance and safety through effective charging techniques is crucial for extending their lifespan, ensuring reliability, and promoting sustainable energy usage. Therefore effective charging techniques are essential for optimizing the performance and longevity of Li-ion batteries. The shortfalls in the design of charging circuit leads to degradation of battery capacity and safety aspects. The charging techniques aim is to prevent overcharging, undercharging, and excessive heat generation[1-10].

According to the state of art, the charging methods existing in the literature for the lithium-ion battery packs, includes traditional type, non-feedback-type, feedback type, and intelligent charging types[10]. In order to ensure that a Li-ion battery charges quickly, a limited feedback control method

built on a simplified electrochemical-thermal model was suggested [11]. The basic concepts underlying the modeling and simulation of various charging protocols, including constant current, constant voltage, multistage constant current, varying current method, and pulse charging methods, are provided [10-21]. Battery life, charging efficiency, and charging time, constant current-constant voltage charging strategy is regarded as the industry standard [22]. Constant current, constant voltage charging has been a popular method for battery charging in electric vehicle applications [23]. A battery monitoring system includes Li-ion battery State of Health (SoH) estimation. The internal resistance and battery capacity are specifically used as SoH validation markers. Due to the difficulty in achieving complete charging and discharging in the above work [23], the constant voltage charging stage, the constant current drop time, constant current drop capacity, and maximum constant current drop rate are used as health indicators [1-4]. A Li-ion battery charging approach using variable current makes more sense. To segment the charging steps according to SoC, the characteristics of the battery will change with SoC [3]. The automatic changeover from constant voltage to constant current using a linear-based battery charger using a SoC technique to charge low-power wearable or implanted medical equipment achieves a charging current from 500 μ A to 20 mA [20-26]. Fast and efficient mitigation of battery string inconsistencies can be achieved through constant current equalization [27].

Pulse charging involves delivering short bursts of high-current pulses to the battery. This technique reduces charging time and improve battery performance. However, pulse charging requires sophisticated control algorithms to optimize pulse parameters and prevent overheating and overcharging [2]. Trickle charging involves delivering a low current to the battery continuously, typically at a rate much lower than its nominal capacity. This technique is usually applicable for battery maintenance, where batteries are stored for long

periods. But prevents batteries from becoming fully discharged, which can lead to capacity loss and reduced lifespan [1-15]. Fast charging circuits aim to reduce charging time while maintaining battery performance and safety. This technique involves higher charging currents and voltages than conventional CC/CV charging, along with advanced thermal management systems to dissipate heat generated during charging [2]. Wireless charging, also known as inductive charging, eliminates the need for physical connectors by transferring power wirelessly from a charging pad to the battery. This offers convenience and flexibility. However, this method must overcome heat dissipation challenges to ensure optimal charging performance and safety [20-30].

Hence, a combined CC-CV charging technique is being presented in this paper. This circuitry aims to offer unparalleled benefits of optimal battery performance, safety, and longevity. The significance of the proposed circuitry lies in delivering a consistent flow of either current or voltage to a load, irrespective of variations in load characteristics or external conditions. In the initial constant current phase, a regulated current is supplied to the battery, preventing overcurrent and ensuring a controlled flow of energy, especially during lower states of charge. On reaching a predetermined state of charge, the mode of operation shifts to CV charging. Hence, a stable voltage is maintained across the battery terminals, allowing a controlled reduction in charging current and preventing overvoltage, thereby safeguarding the battery during the final stages of the charging process.

By understanding and implementing combined control of CC-CV charging method, this paper focuses on understanding the basics to advanced concepts in Li-ion battery charging concepts. The following sections deals with design, simulation circuit, hardware implementation and use of ML for estimating Li-ion batteries.

II. DESIGN AND IMPLEMENTATION OF COMBINED CC-CV CHARGING CIRCUIT

A. Study of CC-CV charging circuit

The generalized block diagram of the described methodology is illustrated in Fig. 1. A single phase, 230V/50Hz AC supply is given to the rectifier circuit with capacitive filter. The processed supply is then stepped down by the buck converter. The converter provides an dc output voltage of 24V. In order to illustrate combined CC-CV charging technique, a buck converter is adapted for studying and implementing for the lithium-ion battery.

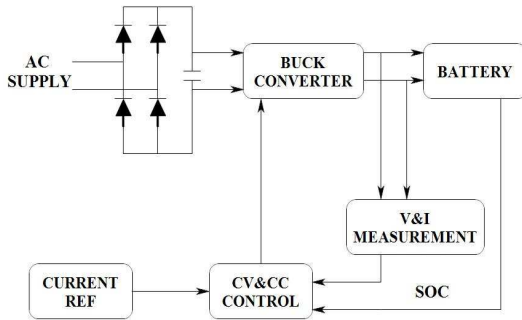


Fig. 1. Generalized block diagram of Li-ion battery charging circuit

The CC-CV controller is used to switch the charging mode from CC to CV after the SoC is achieved. The controller is used to charge the battery with constant current up to the SoC level and then it charges the remaining capacity of battery by constant voltage. Current reference of 10A is given and the reference voltage is taken as the reference voltage at the point of the SoC. Accordingly, the proportional-integral (PI) controller is tuned by the machine learning (ML) algorithm. The algorithm optimizes the PI controller using ML technique, which ensures constant current and constant voltage during charging profile leading to higher efficiency.

B. Design considerations of Buck converter and Li-ion battery

The design equations of the buck converter are considered for the implementation of the Li-ion battery charging system [2-4]. By assuming the efficiency of switch to be nearly 95%. The duty cycle(D) is given by:

$$D = \frac{V_o}{(V_{in} * \eta)} \quad (1)$$

Inductance ripple current (ΔI_L) is assumed to be in the range of 10% to 20% of the output current. The maximum output current is achieved by choosing optimal value of inductor. The inductor value(L) is calculated using (2).

$$L = DT_s * \frac{(V_d - V_o)}{\Delta I_L} \quad (2)$$

The output capacitor (C_{out}) value is selected from (3)

$$C_{out} = \frac{(1-D) \times V_{out} \times (T_s^2)}{(8L \times \Delta V_{out})} \quad (3)$$

From the (1), (2) and (3), the design values of buck are tabulated in the Table I.

TABLE I. Design values of the charging circuit (Buck)

Sl.No	Parameters	Values
1.	Duty Cycle, D	0.6
2.	Switching Frequency, f_{sw}	50kHz
3.	Output Power, P_o	1000 W
4.	Output Voltage, V_o	48V
5.	Inductor, L	2mH
6.	Capacitor, C_{out}	20uF
7.	Ripple in Inductor Current, ΔI_L	0.4 A
8.	Ripple in Capacitor Voltage, ΔV_o	0.48 V

The Li-ion battery parameters selected for the study of combined CC-CV charging circuit is specified in Table II.

TABLE II. Specifications of Li-Ion Battery

Sl.No	Parameters	Values
1.	Battery (20Ah, 1C)	Li-ion
2.	Rated Voltage	24V
3.	Rated Current	20 A

The rated voltage, current and the maximum power rating are noted down from the data sheet for simulation and hardware implementation.

C. Simulation study of combined CC-CV Li-ion battery

The charging of a lithium-ion battery employing combined CC-CV technique is simulated using Simulink environment. The simulation circuit shown in Fig. 2 offers solution to prove the significance of the designed circuit.

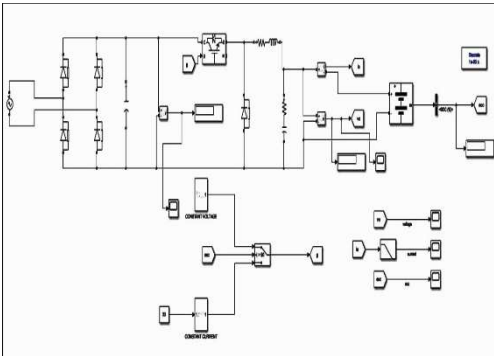


Fig. 2. Simulation of combined CC-CV circuit

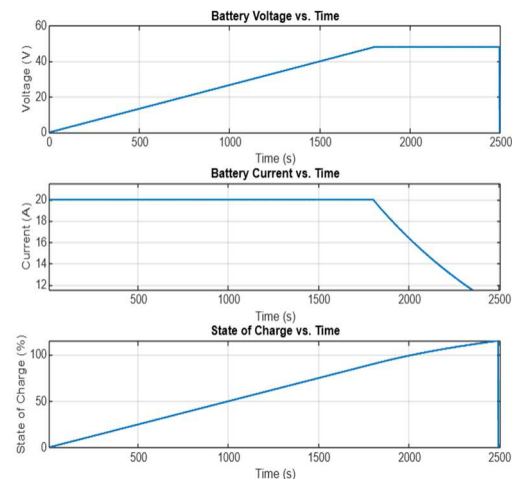


Fig. 3. CV/CC/SoC region

From Fig. 3, we infer that the voltage is changing until the SOC of the battery is achieved. After the SOC is fixed, the voltage remains constant (i.e the battery is charged by the constant voltage). This graph represents the CV region.

From Fig. 4, we infer that the current is increasing gradually till the SOC level. This constant current is used to charge the battery till the SOC. This method is the best way of charging the battery. This area is known as the CC zone. After the SOC level is done, the current slowly decreases, thus maintaining the constant voltage. By implementing the CC charging at the initial state, the battery will quickly be charged and CV charging improves the performance, safety and lifespan of Li-ion batteries.

The sample and hold circuit is used to hold the value of the voltage at the point of SoC, which is later taken as reference by the controller. From the initial level of the

battery, the final SoC level is nearly 95% as noticed from Fig. 3.

D. Hardware implementation of combined CC-CV Li-ion battery system

The various components involved in the circuit implementation has being outlined in Fig. 4. The AC mains supply is step down by a transformer, and then rectified to DC using a rectifier-filter, and further processed through a buck converter.

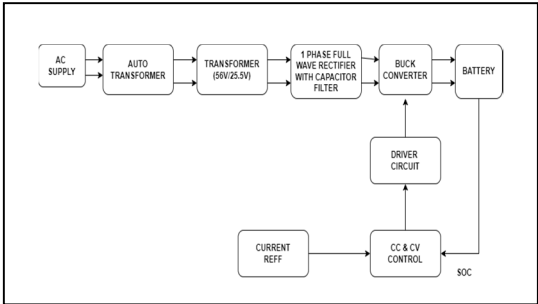


Fig. 4. Outline of CC-CV circuit implementation

The overall hardware implementation circuit along with CC-CV charge controller is shown in the Fig. 5. The charging algorithm first reads the data with the input features, then testing and validating the data. Finally, by testing the model the performance metrics are obtained as displayed in the Table III.

TABLE III. Performance of Li-Ion Battery

Sl.No.	Type	SOC
1	Traditional method	>80%
2	ML based algorithm	>85%

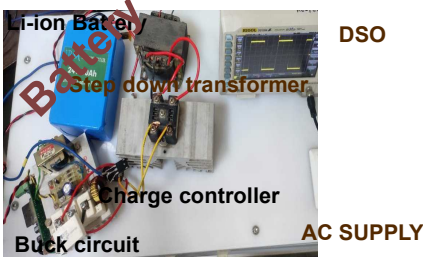


Fig. 5. Hardware implementation of CC-CV charging circuit

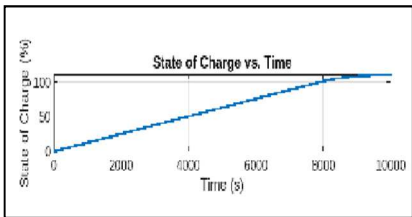


Fig. 6. SoC Vs Time

The controller block then charges the Li-ion battery with a constant-voltage (CV) to obtain final SOC of >85% as noticed

in Fig. 6. The Li-ion battery begins its initial state of charge of 10%. The CC-CV charge controller functions as a constant-current (CC) charging until it reaches the battery voltage of nearly 48 V as represented in Fig. 7.

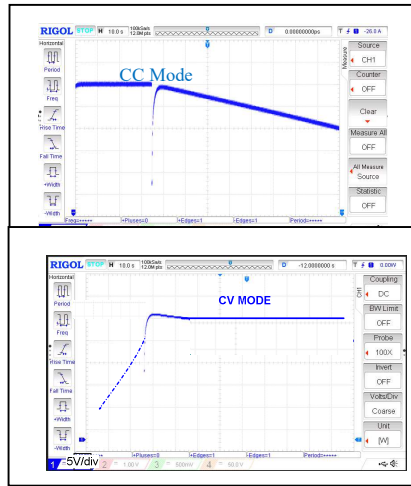


Fig. 7. CC-CV charging profile

III. CONCLUSION

In the CC-CV charging stages, they counterpart each one effectively. Hence, the study of combined charging technique is comparatively better than its individual stages. This work briefs out the simplicity of implementation by incorporating ML algorithm for tuning the PID controllers. The total time required to charge depends on the battery and its charging current. Hence, this study highlights the combined CC-CV charging as a consistent and effective way for Li-ion battery, ensuring real-time safety aspects too.

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